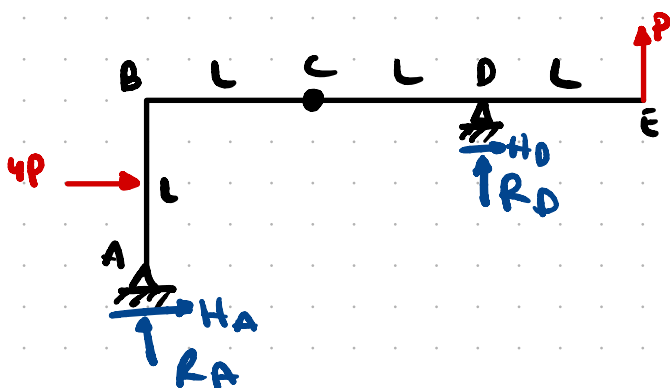




1- Reacciones

2- Leyes

3- Mov. vert. E

Si hay DT en $\overline{AB}$

4- Reacciones

5- Leyes

6- Mov. vert. E

1. $GH = 4 - 3 - 1 = 0$

$$\sum F_H = 0 \rightarrow H_A + H_D = -4P \rightarrow \boxed{H_D = -3P}$$

$$\sum F_V = 0 \rightarrow R_A + R_D = -P \rightarrow \boxed{R_A = P}$$

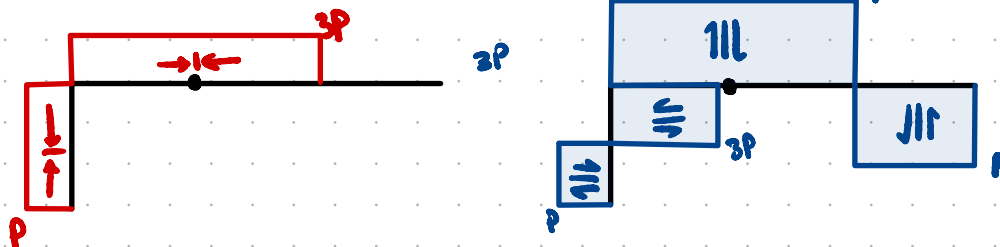
$$\sum M_C^{DCBA} = 0 \rightarrow R_D \cdot L = -P \cdot 2L \rightarrow \boxed{R_D = -2P}$$

$$\sum M_C^{HA} = 0 \rightarrow R_A \cdot L = 4P \cdot \frac{L}{2} + H_A \cdot L \rightarrow \boxed{H_A = -P}$$

2.

AXIL

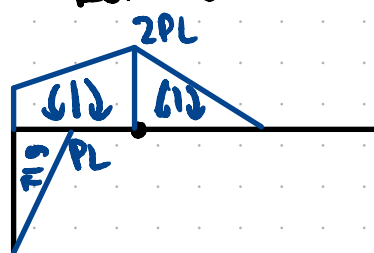
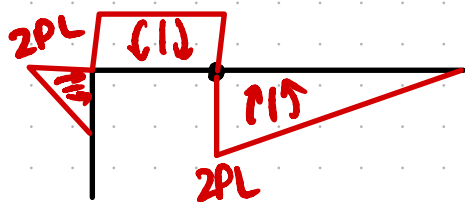
CORTANTE



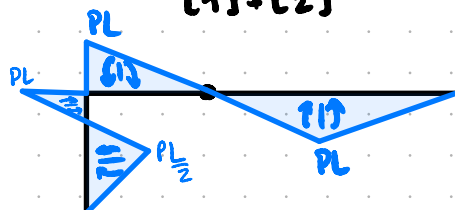
FLECTORES

REACCIONES [1]

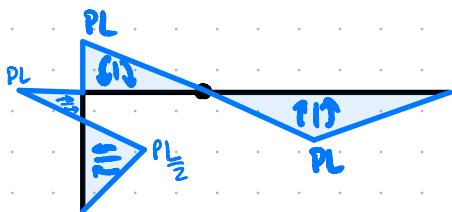
PUEBLAS [1]



[1] + [2]



3-

 $\uparrow V_E?$

$$[1] \uparrow V_E = \tilde{\Theta}_0 \cdot L + \frac{1}{EI} \left[\frac{1}{2} PL \cdot L \cdot \frac{2}{3} L \right]$$

$$\vec{u}_C = \vec{u}_B = 0 = \tilde{\Theta}_A \cdot L + \frac{1}{EI} \left[\frac{1}{2} 2PL \cdot \frac{L}{2} \cdot \frac{L}{6} - \frac{1}{2} PL \cdot L \cdot \frac{1}{3} L \right]$$

$$\rightarrow \tilde{\Theta}_A = \frac{PL^2}{12EI}$$

$$\downarrow V_C^{12a} = \tilde{\Theta}_A \cdot L + \frac{1}{EI} \left[\frac{1}{2} 2PL \cdot \frac{L}{2} \cdot L - \frac{1}{2} PL \cdot L \cdot L + \frac{1}{2} PL \cdot L \cdot \frac{2}{3} L \right]$$

$$\rightarrow \downarrow V_C^{12a} = \frac{PL^3}{12EI} + \frac{PL^3}{3EI} = \frac{5PL^3}{12EI}$$

$$\downarrow V_C^{12a} = \tilde{\Theta}_0 \cdot L + \frac{1}{EI} \left[-\frac{1}{2} PL \cdot L \cdot \frac{2}{3} L \right] = \tilde{\Theta}_0 \cdot L - \frac{PL^3}{3EI}$$

$$\rightarrow \tilde{\Theta}_0 = \frac{3PL^3}{4EI}$$

$$[1] \uparrow V_E = \frac{3PL^3}{4EI} + \frac{PL^3}{3EI} \rightarrow$$

$$\uparrow V_E = \frac{13PL^3}{12EI}$$

4- IGUAL QUE 1

5- IGUAL QUE 2

} Las temperaturas solo producen movimientos, no tensiones

$$6- \downarrow V_C^{12a} = (\dots) - \alpha \cdot \Delta T \cdot L = \frac{5PL^3}{12EI} - \alpha \Delta T L$$

$$\rightarrow \tilde{\Theta}_0 = \frac{3PL^3}{4EI} - \alpha \Delta T L$$

$$\rightarrow \uparrow V_E = \frac{13PL^3}{12EI} - \alpha \Delta T L$$